

# Zengrui Han

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University of Maryland, College Park  
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## APPOINTMENTS

2025–2028     University of Maryland, College Park  
                    Brin Postdoctoral Fellow

2020–2021     University of Science and Technology of China  
                    Research Assistant

## EDUCATION

Ph.D.         Rutgers, the State University of New Jersey,    2020-2025  
                    Advisor: Lev Borisov  
                    Dissertation: GKZ hypergeometric systems and toric mirror symmetry

B.S.         University of Science and Technology of China,    2016-2020

## RESEARCH AREAS

Algebraic Geometry, especially mirror symmetry and related areas.

## PUBLICATIONS AND PREPRINTS

2026         Minimal resolutions of toric substacks by line bundles. arXiv:2604.14384.

2025         Hanlon-Hicks-Lazarev resolution revisited, with Lev Borisov. arXiv:2509.11077. Submitted.  
                    GKZ hypergeometric systems and toric mirror symmetry. Ph.D. Thesis.

2024         Stringy Hodge numbers of Pfaffian double mirrors and Homological Projective Duality.  
                    arXiv:2409.17449. Submitted.  
                    Central charges in local mirror symmetry via hypergeometric duality. **Advances in Mathematics**,  
                    Volume 480, Part B (2025): 110502. arXiv:2404.16258.

2023         Analytic continuation of better-behaved GKZ systems and Fourier-Mukai transforms, **Épjournal de  
Géométrie Algébrique**, Volume 9 (2025), Article no. 11. arXiv:2305.12241  
                    On hypergeometric duality conjecture, with Lev Borisov, **Advances in Mathematics**, Volume 442  
                    (2024): 109582. arXiv:2301.01374

2019         On duality of certain GKZ hypergeometric systems, with Lev Borisov and Chengxi Wang, **Asian  
Journal of Mathematics**, Volume 25 (2021), No.1, 65-88. arXiv:1910.04039

## TALKS AND PRESENTATIONS

- 2026      Geometry Seminar at George Mason University, Apr 2026.  
            AMS Spring Eastern Meeting 2026, Special Session on Group Actions and Combinatorics in Algebraic Geometry and Commutative Algebra, Boston College, Mar 2026.
- 2025      JHU-UMD Algebra and Number Theory Day, University of Maryland, Nov 2025.  
            Syzygies and Mirror Symmetry Virtual Seminar, Feb 2025.
- 2024      Rutgers Algebra Seminar, Rutgers University, Oct 2024  
            Columbia Enumerative Geometry Seminar, Columbia University, Sept 2024  
            Rutgers Algebra Seminar, Rutgers University, April 2024
- 2023      Rutgers Algebra Seminar, Rutgers University, Feb 2023

## SERVICE AND PROFESSIONAL ACTIVITIES

### Seminars

- 2025-      Co-organizer, RIT seminar on Log Geometry, University of Maryland, College Park  
2025-      Co-organizer, Algebraic Geometry Seminar, University of Maryland, College Park

### Refereeing

Referee for Journal of Differential Geometry.

## TEACHING

### University of Maryland

- Math 120: Elementary Calculus I, Fall 2026  
Math 301: Introduction to Mathematical Proof, Spring 2026  
Math 402: Algebraic Structures, Fall 2025

### Rutgers University

- Math 551: Abstract Algebra I, TA, Fall 2024  
Math III: Precalculus, TA, Spring 2024  
Math 552: Abstract Algebra II, TA, Spring 2024  
Math 551: Abstract Algebra I, TA, Fall 2023  
Math 477: Mathematical Theory of Probability, TA, Fall 2023  
Math 244: Differential Equations for Engineering and Physics, TA, Spring 2023  
Math 354: Linear Optimization, TA, Fall 2022  
Math 251: Multivariable Calculus, TA, Spring 2022  
Math 251H: Multivariable Calculus, TA, Fall 2021

## **AWARDS AND HONORS**

|      |                                                                                              |
|------|----------------------------------------------------------------------------------------------|
| 2025 | Rutgers SAS Fellowship, Rutgers University, 2025                                             |
| 2024 | Summer PhD student Research Fellowship, Rutgers University, 2024                             |
| 2019 | Huang Yu Memorial Scholarship, University of Science and Technology of China, 2019           |
| 2018 | Outstanding Student Scholarship, Silver Award, University of Science and Technology of China |
| 2017 | Outstanding Student Scholarship, Silver Award, University of Science and Technology of China |
| 2015 | Chinese Mathematical Olympiad (CMO), Bronze Medal, 2015                                      |

Updated May 2026